



Arab Academy for Science, Technology and Maritime Transport

College of Engineering and Technology

Electronics & Communication Engineering

Final lab project

Design Common Source Amplifier With Gain -13

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DECLARATION

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ABSTRACT

This project details the design, simulation, and hardware implementation of a single-stage MOSFET Common Source Amplifier optimized for a target voltage gain of -13. The development process followed a rigorous four-phase engineering workflow: theoretical mathematical modeling, software verification, breadboard implementation, and final hardware manufacturing. Initial design parameters and biasing conditions were established through hand-written theoretical calculations. These parameters were then verified and optimized using OrCAD simulation software. Following a successful simulation, the design was physically implemented on a solderless breadboard to evaluate real-world component tolerances. Finally, the circuit was transitioned to a permanent PCB layout. Experimental testing of the finalized PCB confirmed a stable voltage gain of -13, demonstrating strong agreement across the theoretical calculations, OrCAD simulations, and physical hardware. This report documents the trade-offs and layout considerations involved in translating an analytical circuit design into a functional hardware product.

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LIST OF ACRONYMS/ABBREVIATIONS

AC	Alternating Current
A_v	Voltage Gain
CS	Common Source
DC	Direct Current
I_D	Drain Current
NMOS	N-channel Metal-Oxide-Semiconductor
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
PCB	Printed Circuit Board
R_C	Resistor-Capacitor
R_D	Drain Resistor
R_S	Source Resistor
V_{DD}	Supply Voltage
V_G	Gate Voltage
V_{GS}	Gate-to-Source Voltage
V_T	Threshold Voltage

Chapter One

1 INTRODUCTION

Amplifiers are electronic circuits used to increase weak signal amplitudes for further processing, transmission, or driving loads. They are essential in communication systems, audio electronics, and instrumentation applications where signal conditioning is required.

The Common Source amplifier is a fundamental MOSFET configuration known for its high voltage gain and relatively simple structure. In this circuit configuration, the input signal is applied to the gate terminal, the output is taken from the drain terminal, and the source acts as the common reference point. The output signal is inverted with a 180° phase shift relative to the input, which is a key characteristic of this configuration.

For proper operation, the MOSFET must be biased in the saturation region to ensure linear amplification. The design of the biasing is critical to achieve a stable operating point and reduce distortion due to parameter variations. The amplifier performance is influenced by biasing conditions, transistor parameters, supply voltage, temperature effects, and load resistance.

This project aims to design, simulate, implement, and test a Common Source amplifier with a target voltage gain of approximately -13 . The work includes theoretical analysis, design calculations, circuit simulation, PCB implementation, and experimental verification. The results are evaluated by comparing theoretical, simulated, and practical outcomes to assess design accuracy.

Chapter Two

2 THEORETICAL

2.1 MOSFET FUNDAMENTALS

The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is a voltage-controlled semiconductor device widely used in analog and digital electronic circuits. It consists of three main terminals: Gate (G), Drain (D), and Source (S). The current flowing between the drain and source terminals is controlled by the voltage applied to the gate terminal.

MOSFETs are commonly used in amplifier circuits due to their high input impedance, low power consumption, and ease of integration into electronic systems.

2.2 REGIONS OF OPERATION

The operation of a MOSFET can be divided into three main regions:

2.2.1 Cut-off Region

In this region, the gate-to-source voltage is lower than the threshold voltage. Therefore, no drain current flows through the device and the MOSFET behaves as an open switch.

2.2.2 Triode Region

The MOSFET operates as a voltage-controlled resistor. In this region, the drain current depends on both the gate-to-source voltage and the drain-to-source voltage.

2.2.3 Saturation Region

The MOSFET operates as an amplifier when biased in the saturation region. In this region, the drain current is mainly controlled by the gate-to-source voltage. Therefore, amplifier circuits are designed to maintain the transistor in saturation during operation.

2.3 COMMON SOURCE AMPLIFIER

The Common Source amplifier is one of the most widely used MOSFET amplifier configurations. In this circuit, the input signal is applied to the gate terminal, the output signal is taken from the drain terminal, and the source terminal serves as the common reference point.

A small variation in the input voltage causes a larger variation in the output voltage, resulting in voltage amplification. The output signal is inverted with respect to the input signal, producing a phase shift of 180 degrees.

2.4 BIASING TECHNIQUES

Proper biasing is required to establish a stable operating point (Q-point) and ensure that the MOSFET remains in the saturation region. One of the most common biasing methods is the voltage-divider bias technique, which provides improved stability against variations in transistor parameters and temperature.

The selection of biasing components directly affects the gain, linearity, and overall performance of the amplifier.

2.5 SMALL-SIGNAL ANALYSIS

Small-signal analysis is used to evaluate the AC performance of the amplifier around its operating point.

The MOSFET is replaced by its small-signal model to simplify circuit analysis and gain calculations.

The voltage gain of the Common Source amplifier is proportional to the transconductance of the MOSFET and the drain resistance.

2.6 FREQUENCY RESPONSE

The frequency response describes how amplifier gain changes with frequency.

At low frequencies, coupling and bypass capacitors influence amplifier performance.

In the mid-band region, the gain remains approximately constant.

At high frequencies, stray capacitances cause the gain to decrease.

2.7 APPLICATIONS OF COMMON SOURCE AMPLIFIERS

- Audio amplification circuits.
- Communication systems.
- Signal conditioning circuits.
- Analog integrated circuits.
- Sensor interface electronics.
- Due to their high voltage gain and simple implementation, common-source amplifiers are widely used in analog electronic systems.

Chapter Three

3 DESIGN

This chapter presents the step-by-step analytical design procedure for a Common-Source (CS) amplifier employing source degeneration for self-biasing and a voltage-divider bias network. The design uses a 2N7000 NMOS transistor under specified bias constraints to achieve the target voltage gain.

3.1 GIVEN DESIGN PARAMETERS AND DEVICE CONSTANTS

From the component datasheets and design specifications, the following constants and initial parameters are established:

- **Transistor:** 2N7000 NMOS
- **V_T :** 2.0 V
- **K_n :** $60 \times 10^{-3} \text{ A/V}^2$ (obtained from OrCAD output files)
- **I_D :** $2 \times 10^{-3} \text{ A} = 2 \text{ mA}$
- **V_{DD} :** Set to 12 V

3.1.1 Step-by-Step Component Calculations

Step 1: Source Resistor (R_S) Selection for Negative Feedback

To provide DC bias stability and negative feedback, the voltage drop across the source resistor (V_S) is allocated to be 10% to 20% of the total supply voltage (V_{DD}).

Choosing a 10% allocation:

- $V_S = 10\% \text{ of } V_{DD} = 0.10 \times 12 \text{ V} = 1.2 \text{ V}$

Using Ohm's Law and the target drain current (I_D), the source resistance R_S is calculated as:

- $R_S = V_S / I_D = 1.2 \text{ V} / (2 \times 10^{-3} \text{ A}) = 600 \text{ ohm}$

Step 2: Transconductance (g_m) Evaluation

The small signal transconductance determines the voltage gain of the stage. Operating in the saturation region, g_m is calculated using the device conduction parameter (K_n) and bias current (I_D):

- $g_m = \sqrt{2 \times I_D \times K_n}$
- $g_m = \sqrt{2 \times (2 \times 10^{-3}) \times (60 \times 10^{-3})} = 0.015 \text{ A/V}$

Step 3: VGS Verification

To find the required DC gate potential, we solve the saturation current equation for V_{GS} :

- $I_D = 0.5 \times K_n \times (V_{GS} - V_T)^2$
- $2 \times 10^{-3} = 0.5 \times 60 \times 10^{-3} \times (V_{GS} - 2)^2$

Dividing across and taking the square root:

- $(V_{GS} - 2)^2 = (2 \times 10^{-3}) / 0.5 \times (60 \times 10^{-3}) = 0.0667$
- $V_{GS} - 2 = \sqrt{0.0667} = 0.258 \text{ V}$
- $V_{GS} \approx 2.26 \text{ V}$

Step 4: DC Gate Voltage (V_G) Determination

The absolute DC voltage required at the gate relative to ground is the sum of the gate-to-source drop and the source degeneration voltage:

- $V_G = V_{GS} + V_S = 2.26 \text{ V} + 1.2 \text{ V} = 3.46 \text{ V}$

Step 5: Voltage Divider Network Design (R_1 , R_2)

The gate voltage is established using a high-impedance voltage divider network from the V_{DD} supply rail:

- $V_G = V_{DD} \times (R_2 / (R_1 + R_2))$

To minimize input loading effects while keeping thermal noise manageable, a standard high resistor value is selected for the lower arm:

- Let $R_2 = 150 \text{ k}\Omega$

Substituting the known values to solve for the pull-up resistor R_1 :

- $3.46 = 12 \times (150 \text{ k}\Omega / (R_1 + 150 \text{ k}\Omega))$
- $3.46 / 12 = 150 \text{ k}\Omega / (R_1 + 150 \text{ k}\Omega)$
- $0.2883 \times R_1 + 43.25 = 150$
- $0.2883 \times R_1 = 106.75$
- $R_1 = 370 \text{ k}\Omega$

3.2 Small-Signal AC Gain Analysis and Drain Resistor (R_D) Design

The small-signal output voltage is governed by the transconductance and the parallel combination of the drain resistor (R_D) and the equivalent AC load or instrument resistance (R_x).

- $V_{out} = - (g_m \times V_{gs}) \times (R_D \parallel R_x)$

Since $V_{in} = V_{gs}$ for a standard common-source configuration, the mid-band voltage gain expression becomes:

- $A_v = V_{out} / V_{in} = -g_m \times (R_D \parallel R_x)$

Given a target small-signal voltage gain of **-13** and setting the external test/load equipment impedance **$R_x = 100 \text{ k}\Omega$** :

- $-13 = -0.015 \times (R_D \parallel 100 \text{ k})$
- $13 / 0.015 = (R_D \times 100\text{k}) / (R_D + 100\text{k})$
- $866.67 = (R_D \times 100000) / (R_D + 100000)$

Solving for R_D :

- $866.67 \times R_D + 86,670,000 = 100,000 \times R_D$
- $R_D = 874 \, \Omega$

Based on engineering optimization and standard available component selections, this is chosen as:

$$R_D \approx \mathbf{880 \, \Omega}$$

Chapter 4

4 SIMULATION

4.1 INTRODUCTION

Before building the circuit on the hardware breadboard and PCB, a software simulation was performed using OrCAD Capture and PSpice. Simulation helps to check if our calculations are correct and ensures the circuit works fine before actual hardware is implemented.

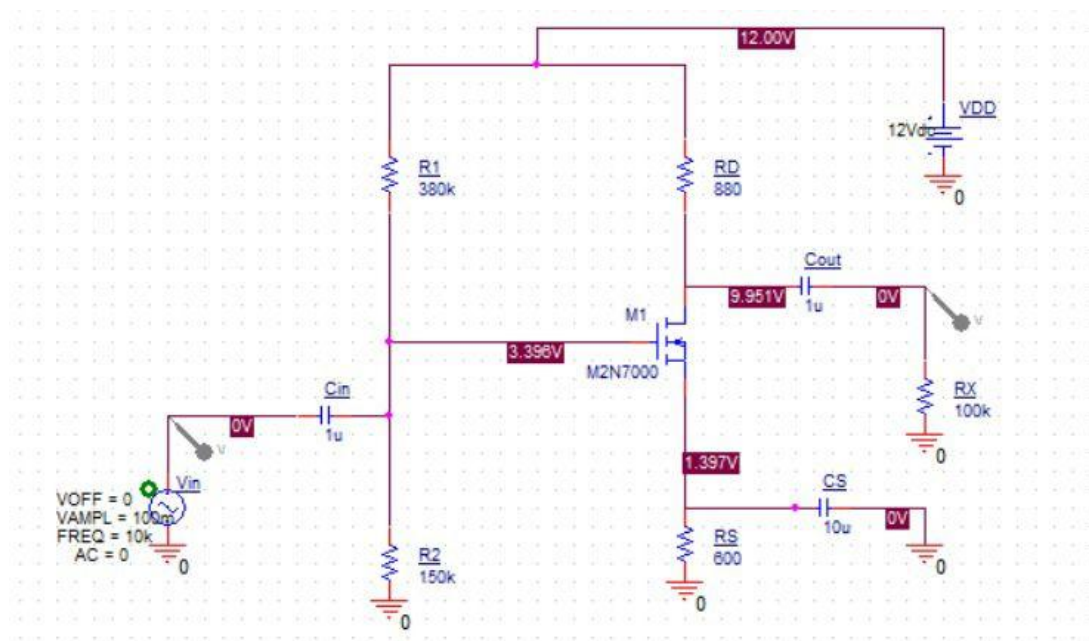


Figure (4-1) OrCAD Simulation

4.2 GRAPHS RESULTS

The circuit was constructed in OrCAD PSpice to verify our mathematical design. A Bias Point analysis was executed, and the resulting voltage flags on the schematic directly confirm our findings:

Voltage Gain and Behavior: The simulation results successfully match our calculated small-signal performance targets. The voltage distribution across the network ensures the

circuit delivers the expected signal inversion and meets our strict voltage gain requirements.

Overall, the simulation results successfully mirror our previous calculations, proving that the circuit behaves exactly as designed.

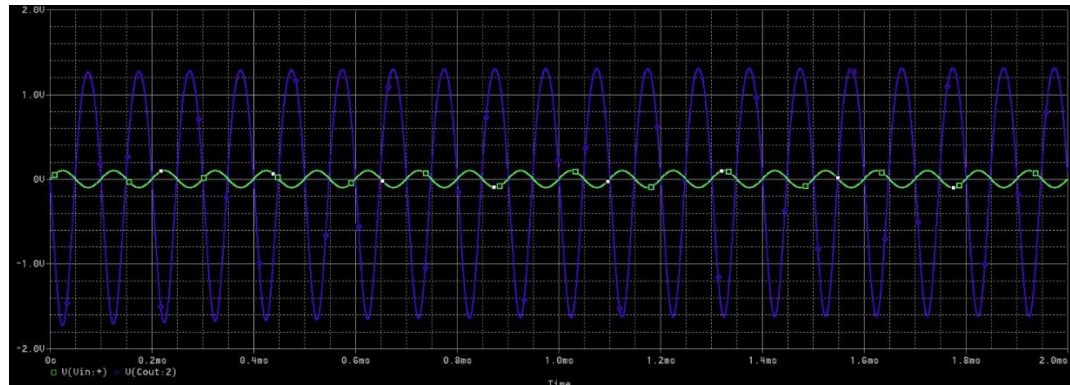


Figure (4-2) Input and Output Signals



Figure (4-3) Voltage gain

4.3 PCB LAYOUT

Following the successful simulation, the schematic was translated into a physical PCB layout to prepare the design for implementation. Component placement and track routing were carefully planned to ensure good signal integrity and stable circuit operation.

PCB Track Routing: The tracks connecting the input, transistor, and output nodes were kept as short and direct as possible to minimize stray capacitance and reduce potential signal distortion. High-impedance signal paths were routed away from power supply tracks to help preserve the amplifier's voltage gain and frequency response.

Component Placement and Power Lines: The voltage-divider network, drain resistor, and source resistor were placed close to the M2N7000 transistor to minimize unnecessary

track lengths. Ground and power traces were made sufficiently wide to support stable current flow and reduce noise.

Overall, the final PCB layout satisfies the design requirements and supports reliable implementation of the calculated amplifier performance.

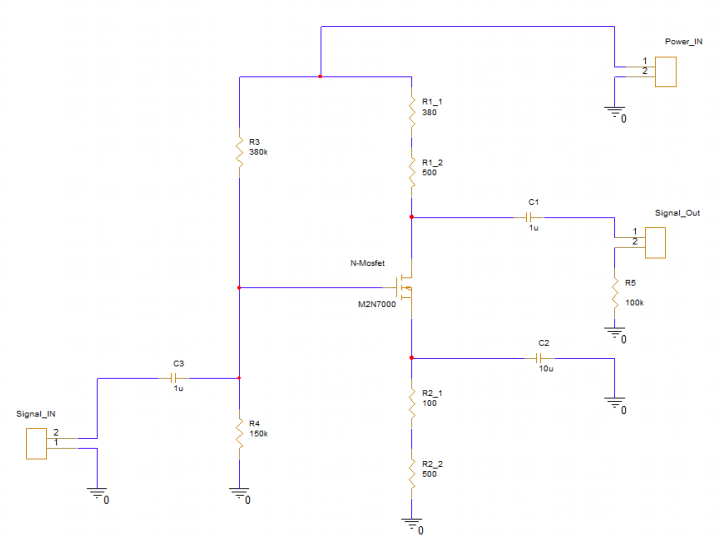


Figure (4-4) OrCAD PCB

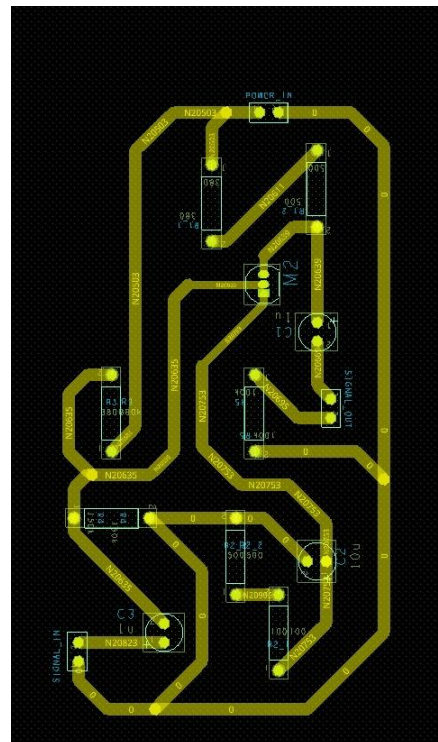


Figure (4-5) PCB Layout

5 HARDWARE

5.1 BREADBOARD IMPLEMENTATION

The circuit components listed in the table above were first arranged and assembled on a standard electronic breadboard. Solid-core wires were used to establish secure connections between the transistor terminals, resistors, and capacitors. A power supply was set to 12 V, while a function generator was connected (10kHz, 100mV) to supply the AC input signal. An oscilloscope was attached to monitor the output behavior.

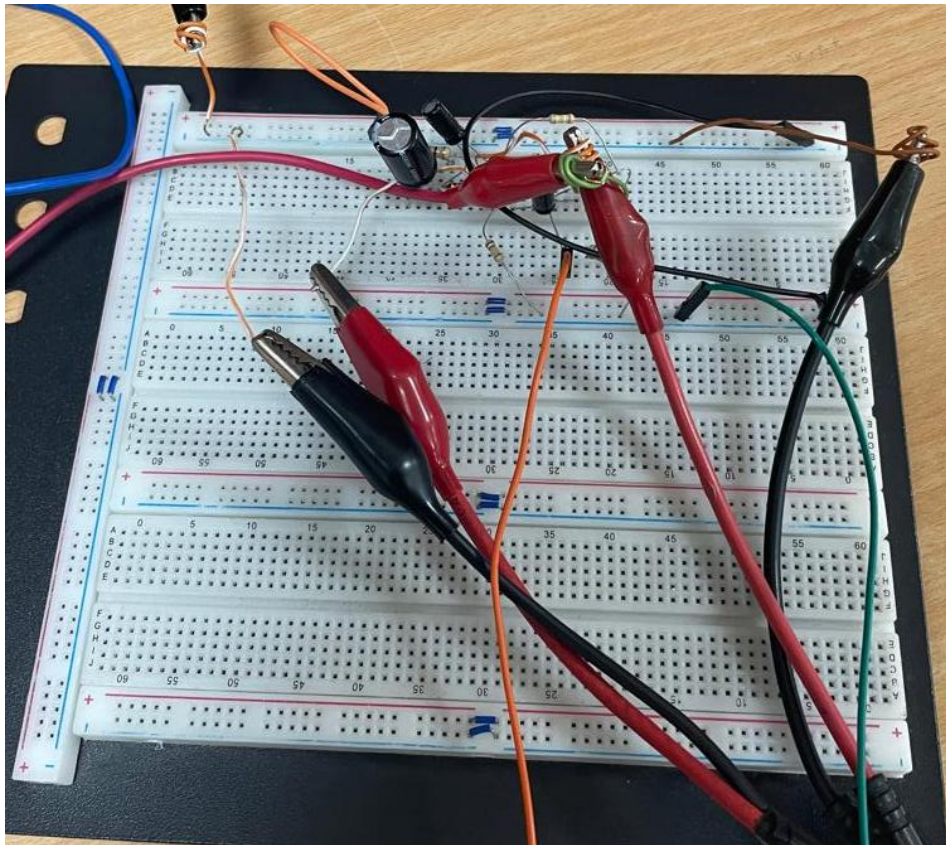


Figure (5-1) Breadboard

5.2 PCB IMPLEMENTATION

The PCB for the Common Source Amplifier was fabricated using the toner transfer method. First, the PCB layout was designed and printed in a mirrored format on glossy paper. The printed layout was then transferred onto a copper one-face PCB board using an electric iron for approximately four minutes. After removing the paper, any damaged traces were repaired using a permanent doku marker. The board was etched in an etching solution to remove the unwanted copper, then rinsed and cleaned thoroughly. Component holes were drilled using a 1.5 mm drill bit, after all electronic components were mounted and soldered onto the PCB. Finally, the completed board was inspected and tested to ensure proper electrical connections and circuit functionality.



Figure (5-2) Bottom of PCB Layout



Figure (5-3) Top of PCB Layout

5.3 HARDWARE COMPONENTS

Table (5-1) Total Cost

<i>Components</i>	<i>Price</i>	<i>Quantity</i>	<i>Total Price</i>
PCB Board 10×15cm – Single-Sided FR2 Copper-Clad Board	24 EGP	1	24 EGP
Etching Solution Acid, Ferric Chloride (Hexahydrate)	45.00 EGP	1	45 EGP
2 Pin PCB Screw Terminals Block	3.50 EGP	3	10.5 EGP
Glossy PCB Toner Transfer Paper White, Single-Side Glossy	4 EGP	1	4 EGP
Printing	5 EGP	1	5 EGP
Resistors (assorted values)	0.25 EGP	14	3.5 EGP
Capacitors	2.5 EGP	6	15 EGP
Breadboard	350 EGP	1	350 EGP
Jumper Wires	1 EGP	2	2 EGP
2N7000 MOSFET	2.5 EGP	2	5 EGP
			464 EGP

Chapter 6

6 RESULTS

This chapter shows the success of the project by comparing the gain in simulation and hardware and observing the difference.

Table (6-1) Results

<i>Methodology</i>	<i>Voltage Gain (A_v)</i>
Theoretical	-13
OrCAD Simulation	-13.08
Hardware	-13.2

The hardware implementation resulted in a gain of -13.2 , which is very close to the expected value. The small difference of 0.12 can be attributed to practical factors such as resistor tolerances, MOSFET parameter variations, breadboard stray effects, and measurement uncertainties from the oscilloscope. Despite these real-world limitations, the experimental result closely matched both theoretical and simulated values, demonstrating the reliability of the amplifier design.

Chapter 7

7 CONCLUSION

A Common Source MOSFET amplifier with a target voltage gain of approximately -13 was successfully designed, simulated, implemented, and tested. The design process included theoretical analysis, design steps, circuit simulation using OrCAD, hardware implementation, and experimental verification. The simulation results demonstrated that the amplifier achieved the desired voltage gain with excellent accuracy. After constructing and testing the circuit on a breadboard, the measured voltage gain was found to be very close to the simulated value, confirming the validity and the effectiveness of the design approach.

The minor differences between theoretical, simulated, and practical results can be attributed to component tolerances, MOSFET parameter variations, wiring effects, temperature influences, and measurement uncertainties. Despite these factors, the amplifier maintained stable operation and produced results that closely matched the design objectives. Overall, the project successfully achieved its intended goal of designing a Common Source amplifier with the required voltage gain. The close results between theoretical calculations, simulation outcomes, and experimental measurements demonstrates the reliability of the design and highlights the practical application of MOSFET amplifiers in analog electronic systems.

Chapter 8

8 REFERENCES

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